بِسْمِ اللّٰهِ الرَّحْمٰنِ الرَّحِيْمِ

اَلْحَمْدُ لِلّٰهِ رَبِّ الْعَالَمِيْنَ

وَالصَّلٰوةُ وَالسَّلَامُ عَلٰى خَاتَمِ النَّبِيِّيْنَ

IPSec

Securing entire IP payload

IP Security (IPsec)

- Various application security mechanisms
  - S/MIME, Kerberos, SSL/HTTPS
- Security concerns cross protocol layers
- Would like security implemented by the network for all applications
- Besides providing security for lower layer (network transport) protocols e.g. routing protocols

IPSec

general IP Security mechanism provides:

- authentication
- confidentiality
- key management

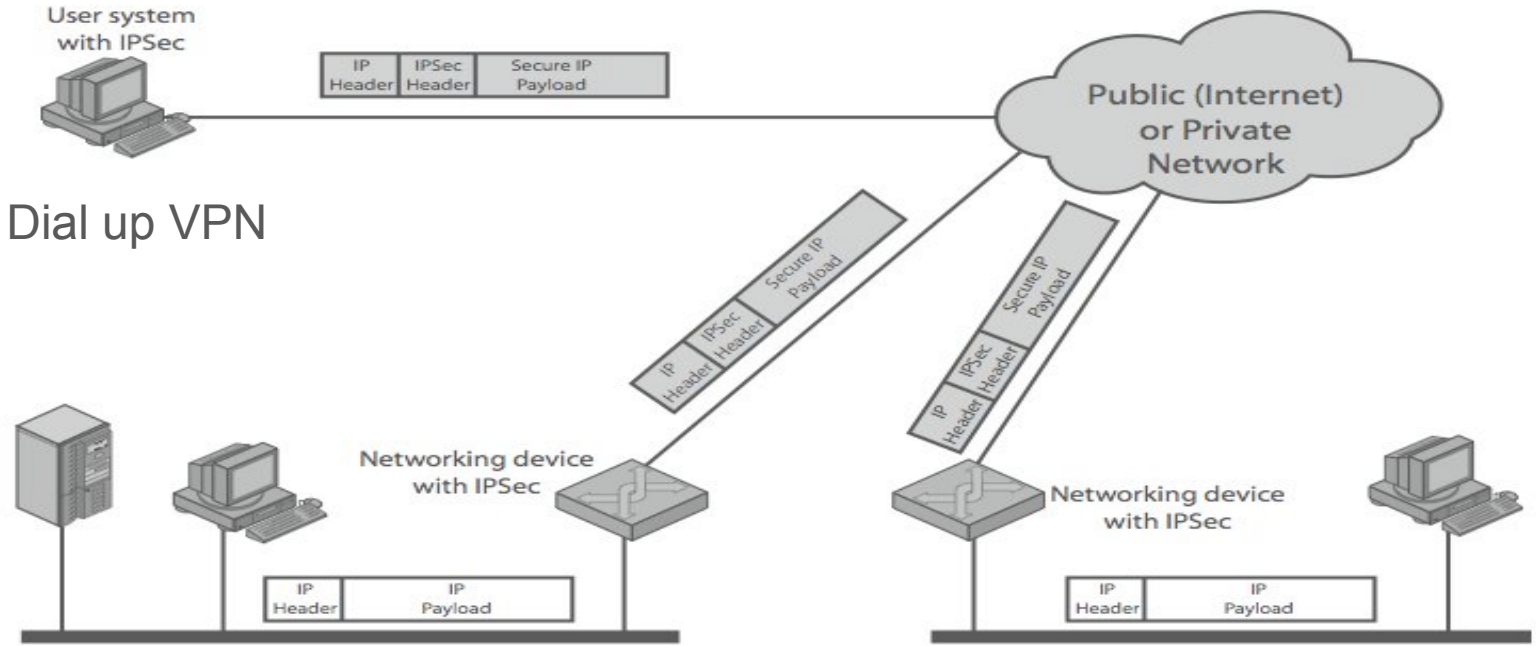
For entire IP payload i.e. any application

Virtual Private Networks VPN

applicable to use over LANs, across public & private WANs, & for the Internet

IPSec Uses

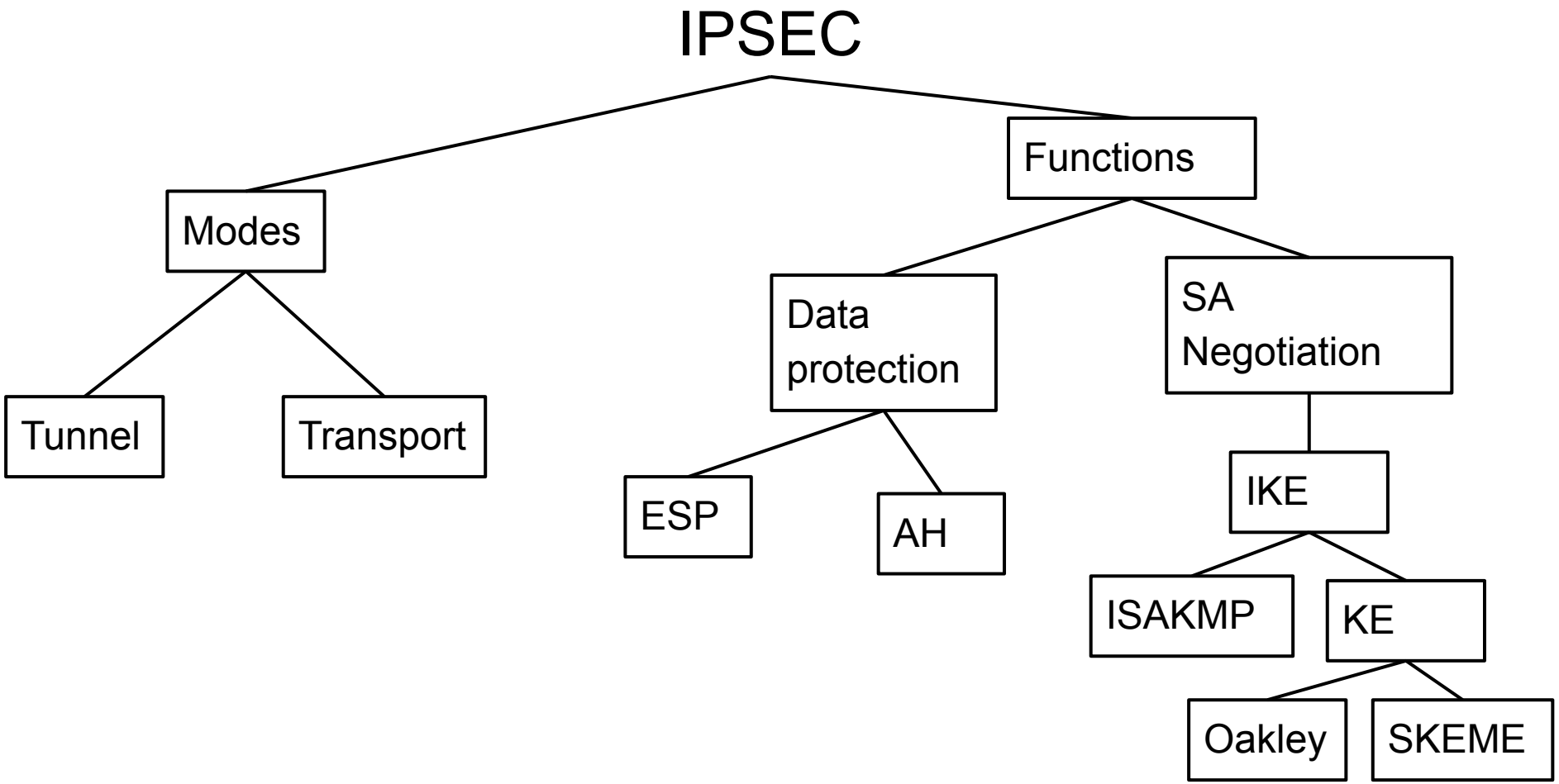
Protects entire IP payload. (any application)



Site to site (point to point) VPN

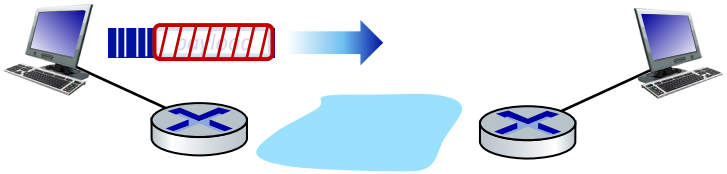
Benefits of IPsec

- When implemented in a firewall or router, it provides strong security to all traffic crossing the perimeter
- In a firewall it is resistant to bypass
- Below transport layer, hence transparent to applications
- Can be transparent to end users
- Can provide security for individual users
- Secures routing architecture


IPSEC
Functions
Modes
Data protection
SA Negotiation
Tunnel
Transport
ESP
AH
IKE
ISAKMP
KE
Oakley
SKEME

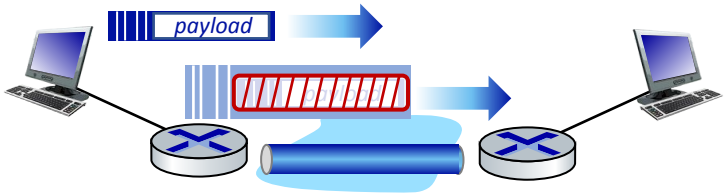
IP Sec

- provides datagram-level encryption, authentication, integrity
  - for both user traffic and control traffic (e.g., BGP, DNS messages)
- two "modes":



transport mode:

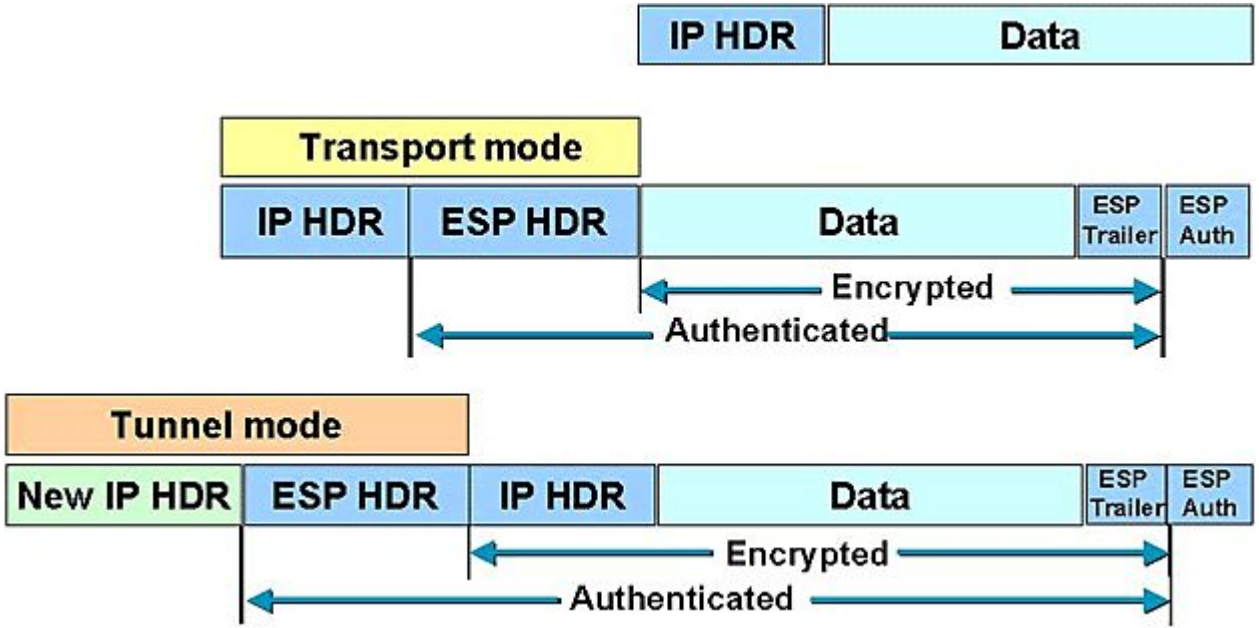
- *only* datagram *payload* is encrypted, authenticated



tunnel mode:

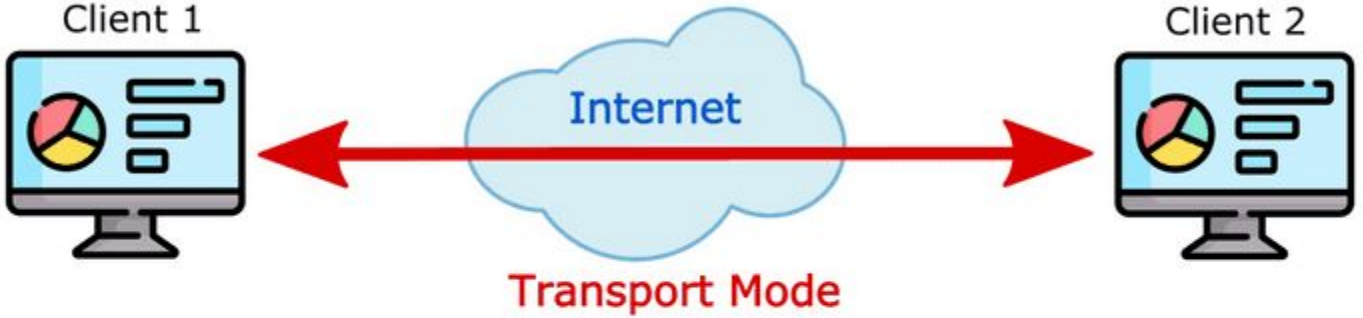
- entire datagram is encrypted, authenticated
- encrypted datagram encapsulated in new datagram with new IP header, tunneled to destination

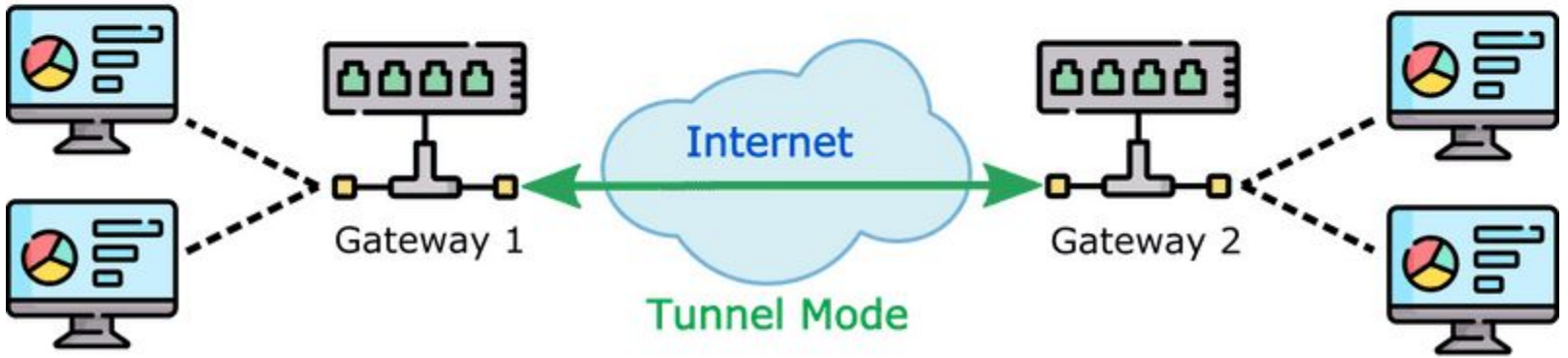
IPSEC modes



https://www.ciscopress.com/articles/article.asp?p=25477

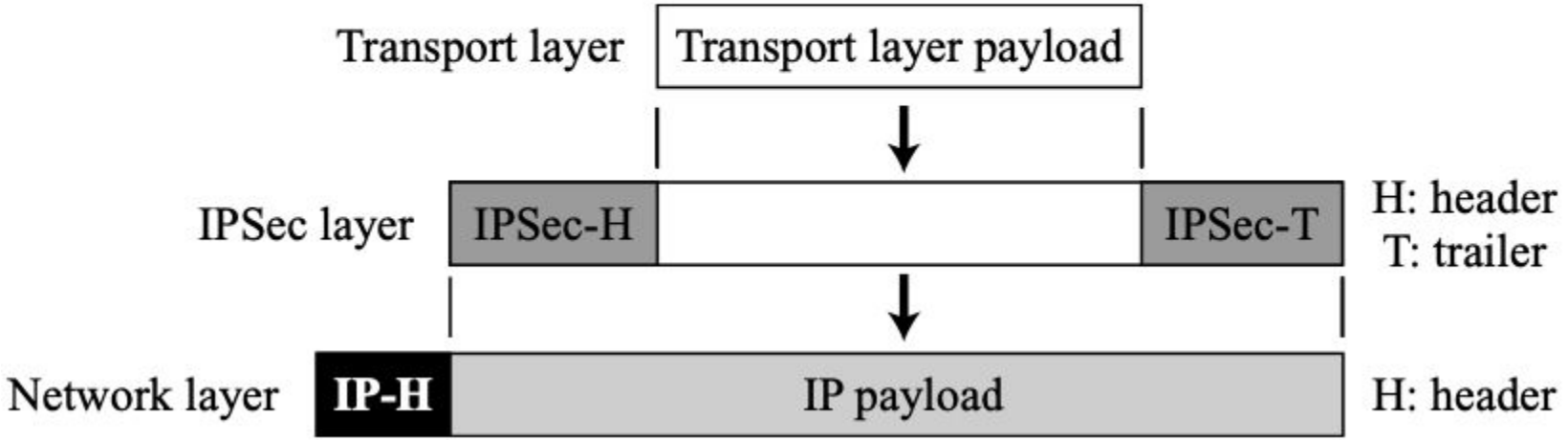
IPSec Modes



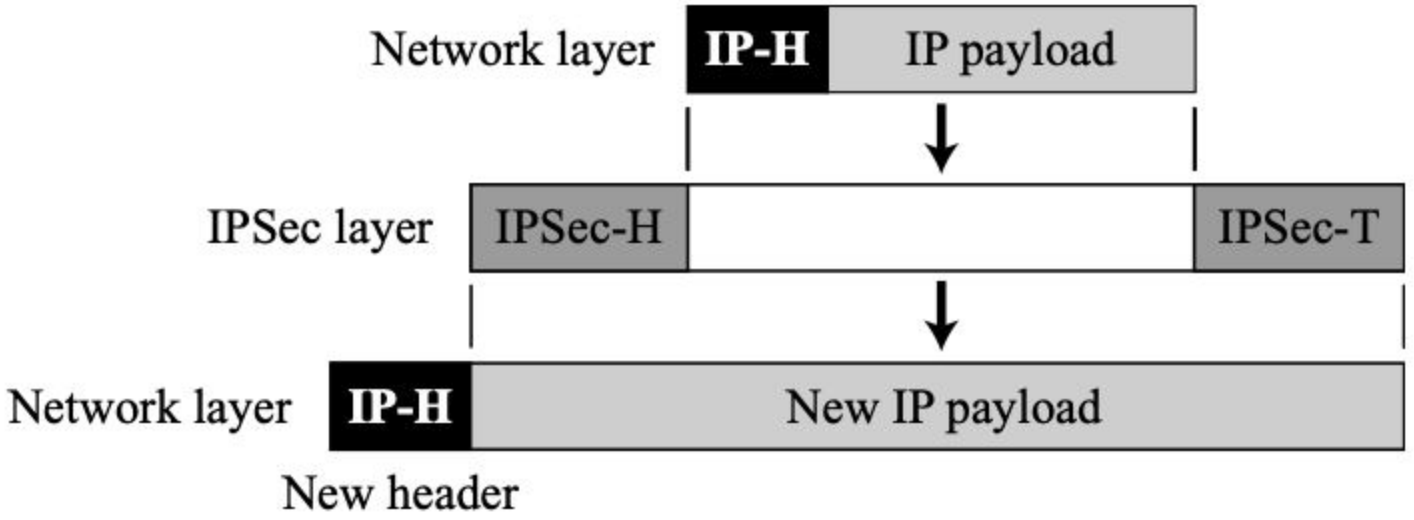


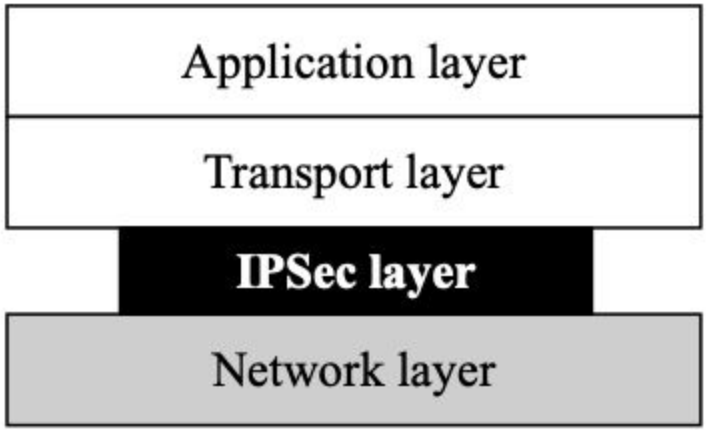
https://www.twingate.com/blog/ipsec-tunnel-mode

.2 *IPSec in transport mode*

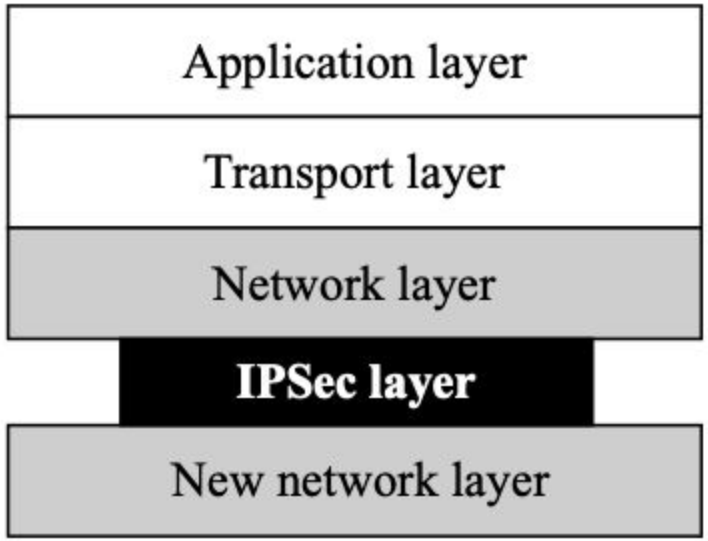


IPSec in tunnel mode





Transport Mode



Tunnel Mode

Transport and Tunnel Modes

Transport Mode

- **Extends to the payload of an IP packet**
- **Typically used for end-to-end communication between two hosts**
- ESP encrypts and optionally authenticates the IP payload but not the IP header

Tunnel Mode

- **Provides protection to the entire IP packet**
- The entire original packet travels through a tunnel from one point of an IP network to another
- **Used when one or both ends of a security association are a security gateway**
- A number of hosts on networks behind firewalls may engage in secure communications without implementing IPsec

IP Security Architecture

Ipsec has two main functions:

1. **Authentication and encryption**
   - Authentication Header (AH) (only authentication)
   - Encapsulating Security Payload (ESP)
2. **Key and Parameters Exchange function**

VPNs want both authentication/encryption hence usually use ESP

The Scope of IPsec

Provides two main functions:

- **A combined authentication/encryption function called Encapsulating Security Payload (ESP)**
- **Key exchange function**

VPNs want both authentication and encryption hence mostly use ESP

Also an authentication-only function, implemented using an Authentication Header (AH)

- **Because message authentication is provided by ESP, the use of AH is included in IPsecv3 for backward compatibility but should not be used in new applications**

IPsec Data protection protocols

Ipsec

Encryption and authentication (ESP/AH)

Key exchange and algorithms negotiations (IKE)

Two IPsec protocols

- Authentication Header (AH) protocol [RFC 4302]
  - provides source authentication & data integrity but *not* confidentiality
  - **to be discontinued included just for compatibility**

- Encapsulation Security Protocol (ESP) [RFC 4303]
  - provides source authentication, data integrity, *and confidentiality*
  - **more widely used than AH**

Figure 18.7 *Authentication Header (AH) protocol*

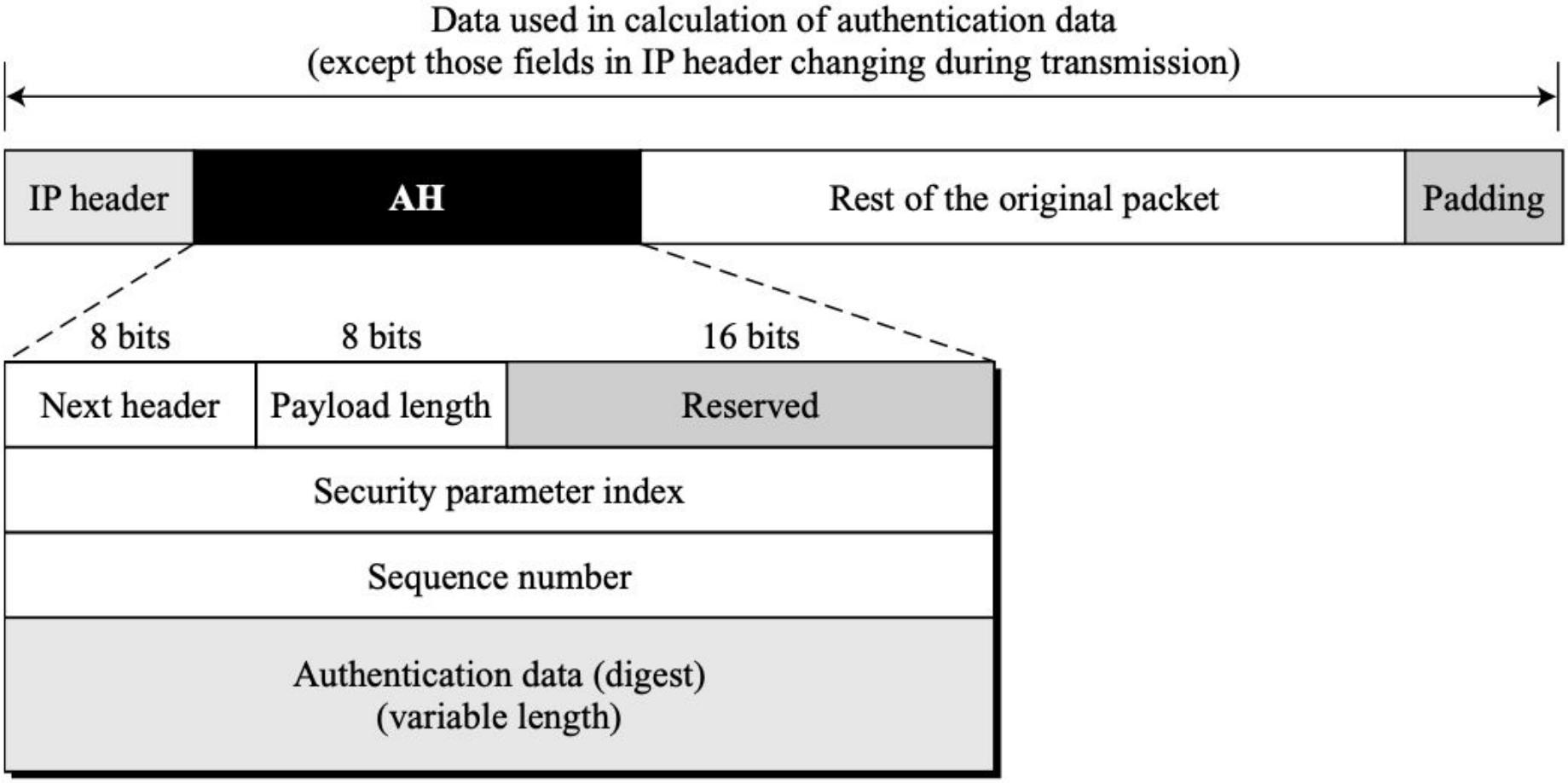
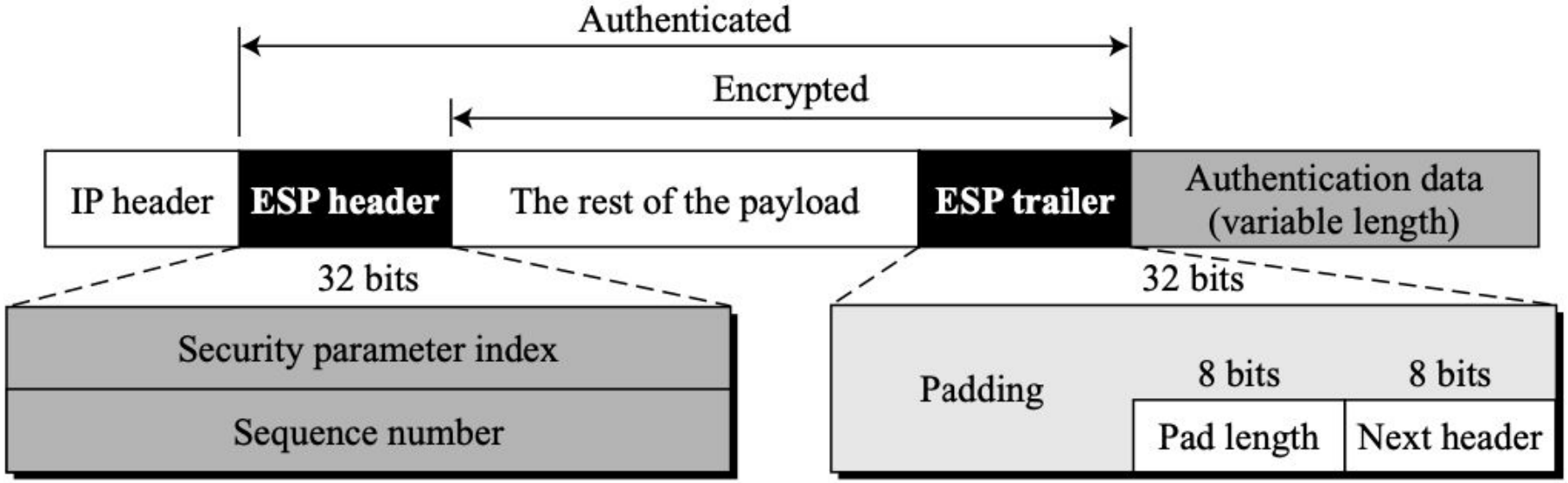


Figure 18.8 *ESP*



Security parameter index. a 32-bit virtual circuit identifier that is the same for all packets sent during a connection called a Security Association between a pair of IPs

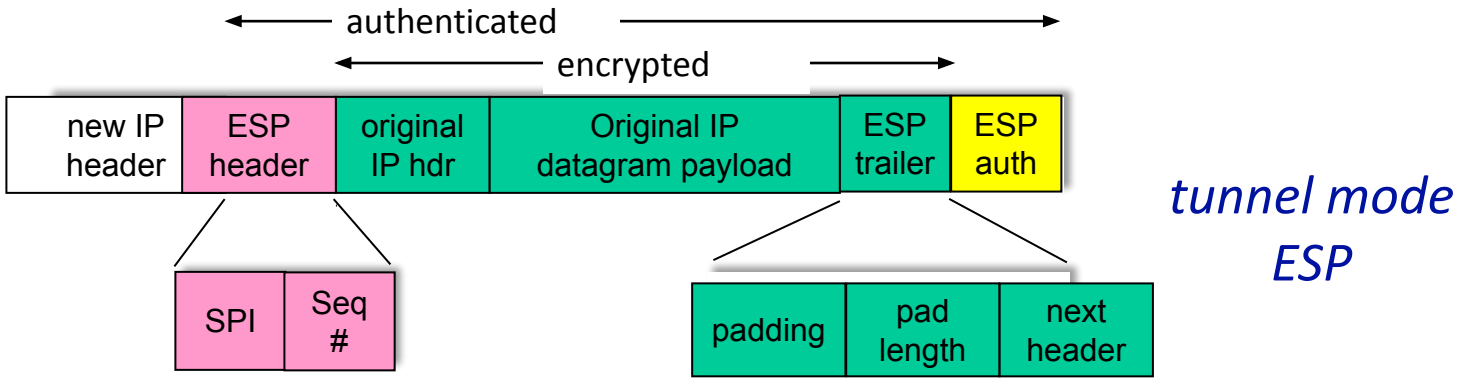
Sequence number. A 32-bit sequence number to provide ordering information for a sequence of datagrams.

The sequence numbers prevent a replay attack and are not repeated even if a packet is retransmitted. A sequence number does not wrap around after it reaches 2^32; rather a new connection must be established.

Next Header: Identifies protocol inside

Authentication data: Digest/Hash value for payload

IPsec datagram



- **ESP trailer:**
  - padding for block ciphers
  - Next header to indicate the protocol within esp header (protocol in ipsec payload)
- **ESP header:**
  - SPI, so receiving entity knows what to do (will explain subsequently)
  - sequence number, to thwart replay attacks (packet discarded if seq # is old)
- **ESP auth:** MAC created with shared secret key

Security Associations

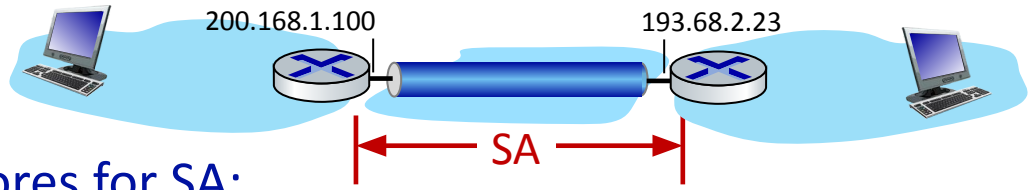
- A one-way relationship between sender and receiver that affords security for traffic flow
  - **If a peer relationship is needed for two-way secure exchange then two security associations are required**
- Is uniquely identified by the Destination Address in the IPv4 or IPv6 header and the SPI in the enclosed extension header (AH or ESP)

Defined by 3 parameters:

- Security Parameter Index (SPI)
- IP Destination Address
- Protocol Identifier (ESP or AH)

Security associations (SAs)

- before sending data, security association (SA) established from sending to receiving entity (unidirectional)
- Sending, receiving entitles maintain *state information* about SA
  - recall: TCP endpoints also maintain state info
  - IP is connectionless; **IPsec is connection-oriented!**



R1 stores for SA:

- 32-bit identifier: *Security Parameter Index (SPI)*
- origin SA interface (200.168.1.100)
- destination SA interface (193.68.2.23)
- type of encryption used
- encryption key
- type of integrity check used
- authentication key

SA DB

< SPI, DA, P >			
< SPI, DA, P >			
< SPI, DA, P >			
< SPI, DA, P >			

Security Association DB

SA Identification

SPI: Security Parameter Index

DA: Destination Address

P: Protocol

SPD Security Policy DB

Index

< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	
< SA, DA, Name, P, SPort, DPort >	

Legend:

SA: Source Address

DA: Destination Address

P: Protocol

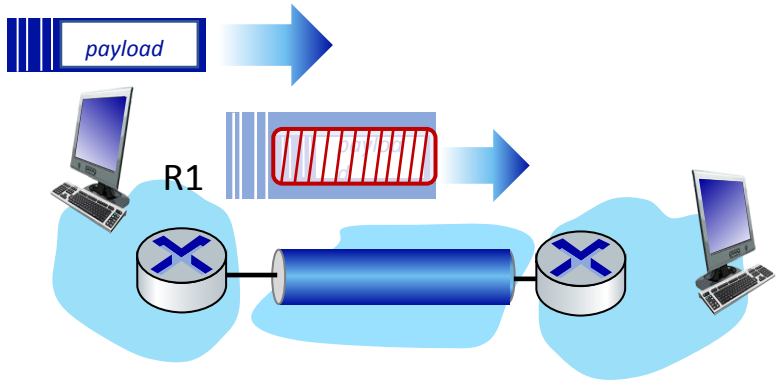
SPort: Source Port

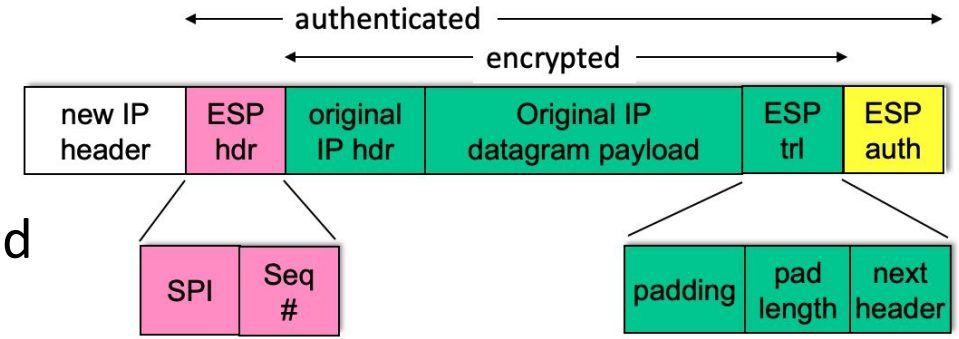
DPort: Destination Port

ESP tunnel mode: actions

at R1:

- appends ESP trailer to original datagram
- encrypts result using algorithm & key specified by SA
- appends ESP header to front of this encrypted quantity
- creates authentication MAC using algorithm and key specified in SA and appends MAC as ESP Auth
- creates new IP header, new IP header fields, addresses to tunnel endpoint





IKE: Internet Key Exchange

- *example:* manual establishment of IPsec SAs in IPsec endpoints:

  Example SA:

  SPI: 12345
  Source IP: 200.168.1.100
  Dest IP: 193.68.2.23
  Protocol: ESP
  Encryption algorithm: 3DES-cbc
  HMAC algorithm: MD5
  Encryption key: 0x7aeaca…
  HMAC key:0xc0291f…

- manual keying is impractical for VPN with 100s of endpoints
- instead use IPsec IKE (Internet Key Exchange)

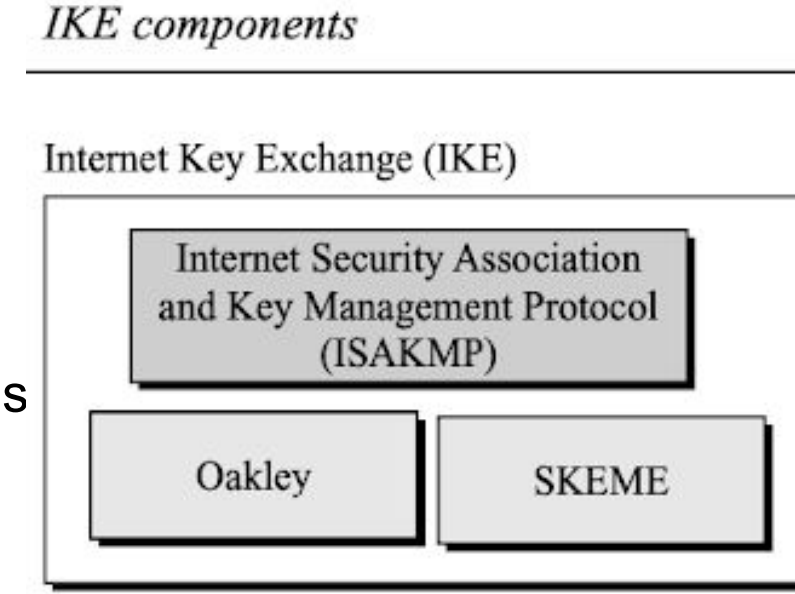
Key Management

- handles key generation & distribution
- typically need 2 pairs of keys
  - 2 per direction for AH & ESP
- manual key management
  - sysadmin manually configures every system
- automated key management
  - automated system for on demand creation of keys for SA's in large systems
  - has Oakley & ISAKMP elements

ISAKMP

Internet Security Association and Key Management Protocol (ISAKMP)

- Negotiates (between both peers) encryption, integrity, authentication algorithms, SPI and key exchange protocols **for a particular SA**
- supports **Oakley** and **SKEME** key exchange protocols



Oakley: Implements key exchange techniques based on the Diffie-Hellman algorithm
SKEME: Public Key based key exchange